

A

B

C

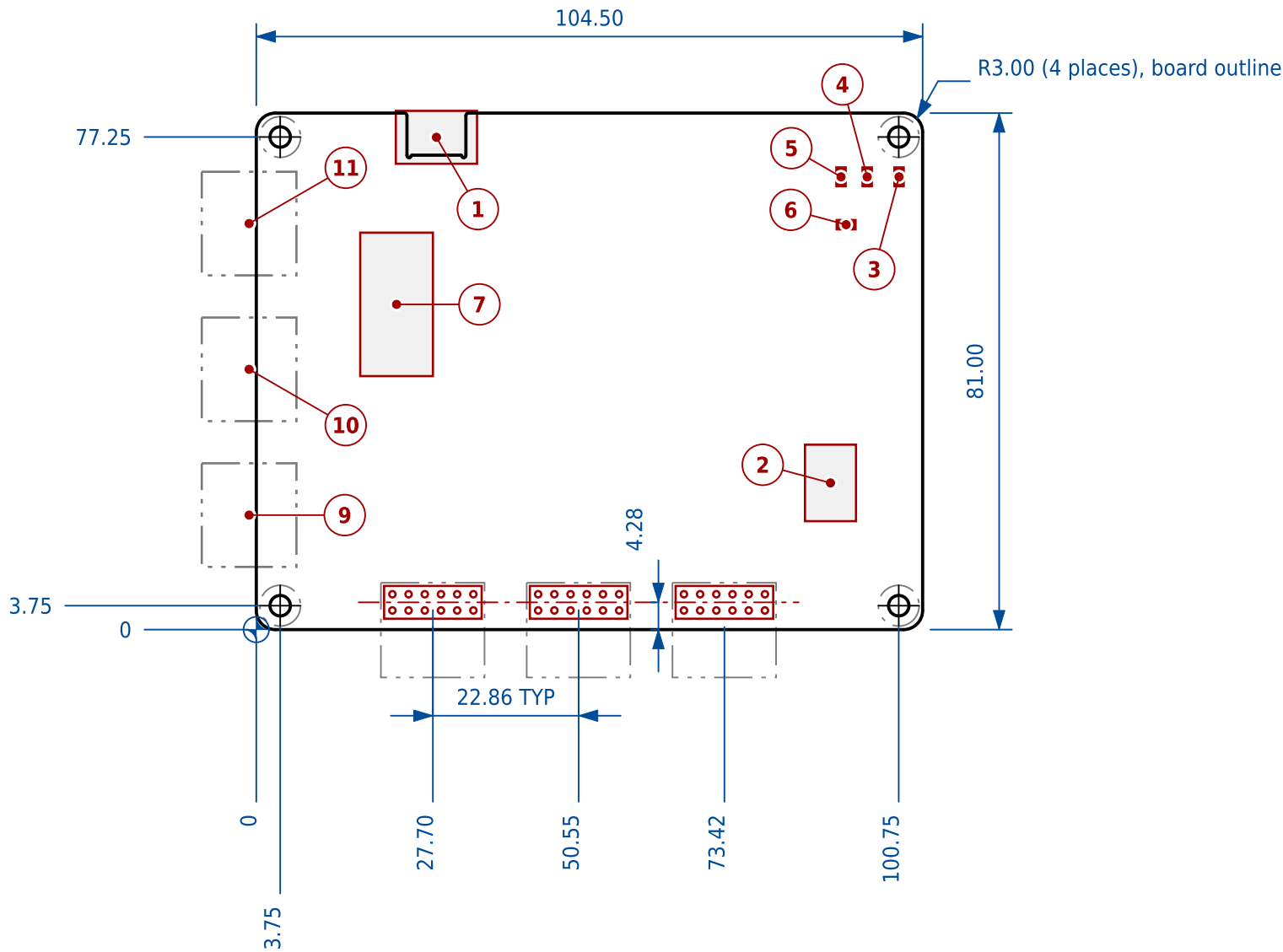
D

A

B

C

D



### NOTES

- Viewed from the component side.
- Ships in TT03p5 ASIC IC kit, TT04 Dev Kit and TT05 Dev Kit.
- Hole IDs are the KiCad reference designators.
- TT03p5's board is recorded as rev 1.2.1 but unconfirmed: the TT03p5 render is labelled v1.1.2.
- v1.2.2c is Tiny Tapeout's name for rev 1.2.3, an assembly variant of 1.2.2.
- The revisions covered are mechanically identical; only the electrical design differs.
- Pmod host pitch, 22.86 mm (0.9 in), is the Digilent Pmod Interface Specification 1.2.0 figure and is the same on every revision; the mounting plate, TT-MP-01, registers against it.
- Assembled envelope, connector overhang included: 104.50 x 81.35 mm.

### NOTES (continued)

- The notch in the upper edge is a recess for the USB-C shell.

### SOURCES

- KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad\_pcb @ 8aad3f8 - title block: TinyTapeout 4+ Demoboard rev 1.2.1, dated 2024-04-12
- KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad\_pcb @ cfdd80d7b - title block: TinyTapeout 4+ Demoboard rev 1.2.2, dated 2024-04-26
- KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad\_pcb @ a88cbc08b - title block: TinyTapeout 4+ Demoboard rev 1.2.3, dated 2024-06-11
- Shuttle mapping: <https://mith.ro/tt-demoboards/> - used by: TT03p5, TT04, TT05

### HOLE SCHEDULE

| ID  | X mm   | Y mm  | DIA mm | KEEPOUT mm |
|-----|--------|-------|--------|------------|
| MT1 | 3.75   | 3.75  | 3.20   | 6.40       |
| MT2 | 100.75 | 3.75  | 3.20   | 6.40       |
| MT3 | 3.75   | 77.25 | 3.20   | 6.40       |
| MT4 | 100.75 | 77.25 | 3.20   | 6.40       |

### PMOD HOST HEADERS

| PORT   | REF | EDGE   | CX mm  | CY mm | PIN1 X mm | PIN1 Y mm |
|--------|-----|--------|--------|-------|-----------|-----------|
| INPUT  | J3  | bottom | 27.695 | 4.275 | 34.045    | 5.545     |
| BIDIR  | J5  | bottom | 50.555 | 4.275 | 56.905    | 5.545     |
| OUTPUT | J6  | bottom | 73.415 | 4.275 | 79.765    | 5.545     |

### FEATURE SCHEDULE

| #  | FEATURE                               | X EXTENT mm      | Y EXTENT mm    |
|----|---------------------------------------|------------------|----------------|
| 1  | USB-C power / control                 | 21.88 to 34.62   | 73.05 to 81.35 |
| 2  | 7-segment display                     | 86.05 to 94.05   | 17.00 to 29.00 |
| 3  | LED D1                                | 100.07 to 101.53 | 69.52 to 72.48 |
| 4  | LED D2                                | 95.07 to 96.53   | 69.52 to 72.48 |
| 5  | LED D3                                | 90.97 to 92.43   | 69.52 to 72.48 |
| 6  | LED D4                                | 91.02 to 93.98   | 62.77 to 64.23 |
| 7  | 8-way input DIP switch                | 16.30 to 27.70   | 39.75 to 62.25 |
| 8  | Second DIP switch - not on this board | -                | -              |
| 9  | Side Pmod J12 (not fitted)            | -8.55 to 6.30    | 9.83 to 26.08  |
| 10 | Side Pmod J13 (not fitted)            | -8.55 to 6.30    | 32.69 to 48.94 |
| 11 | Side Pmod J14 (not fitted)            | -8.55 to 6.30    | 55.55 to 71.80 |

### LEGEND

- Board outline
- Component body, scheduled
- Position not fitted on this revision
- Pmod connector body, hole keep-out
- Dimension, extension and leader

### DB 4+ V1.2.1 / DB 4+ V1.2.2 / DB 4+ V1.2.2C

tinytapeout-demo rev 1.2.1, 1.2.2 and 1.2.3

|                  |                   |                                                |       |
|------------------|-------------------|------------------------------------------------|-------|
| DRAWN            | VERSION           | UNITS                                          | SCALE |
| generated        | v0.0-142-g7459f53 | mm                                             | 1:1   |
| SIZE             | SHEET             | DRAWING NO                                     | REV   |
| A3               | 1 OF 1            | TT-DB-02                                       | A     |
| MATERIAL         |                   | GENERAL TOLERANCE                              |       |
| PCB, 1.6 nominal |                   | edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08 |       |